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NILKAMAL SCHOOL OF MATHEMATICS, APPLIED STATISTICS & ANALYTICS

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Machine Learning vs Traditional Actuarial Underwriting: When Applicants Hide Information

A simulation study, told through the story of one insurance company "Meridian Life".

THE FORM SAYS
₹20,699 / year

≠

THE TRUTH
₹41,774 / year

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Nine sections of the study, five chapters in the life of one insurer

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Problem Statement: Same Person, Two Records

Meridian Life prices what the form says, not what is true.

WHAT THE INSURER SEES

- Age 50, male · BMI 27
- Smoker: No
- Hypertension: No

₹20,699

per year (traditional method)

≠

WHAT IS TRUE

- Age 50, male · BMI 27
- **Smoker: YES (22 pack-years)**
- Hypertension: No

₹41,774

per year (fair price, known from the simulation)

A hidden smoker pays about half the fair price. Who prices this better, and who catches the liars?

Simulated data and illustrative assumptions · results hold for this simulated world

Introduction to Underwriting

Before it sells a policy, Meridian must guess each applicant's 10-year claim chance.

Traditional rule-based

- Mortality table by age and sex
- Fixed multiplier (loading) per risk factor
- Does not learn from new data
- Treats factors separately: may miss compounding

VS

Machine learning

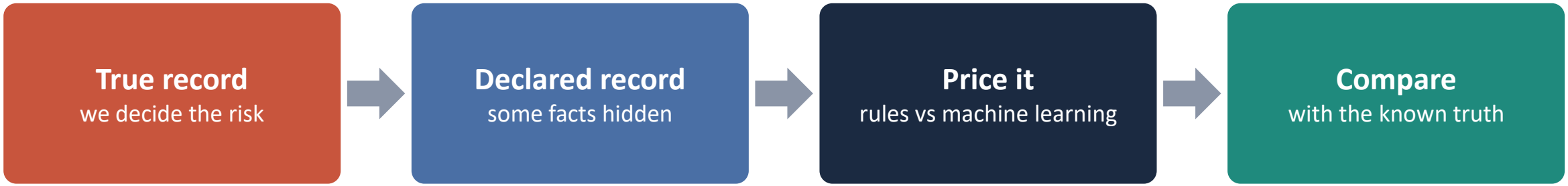
- Trains on data, learns the pattern
- Captures how risk factors strengthen each other
- Needs enough data to beat fixed rules
- Models: GLM, Cox, XGBoost

1 Applicant declares facts → 2 Estimate 10-year claim chance p → 3 Premium = 1.25 × p × SA ÷ T → 4 Standard · Rated · Decline

Simulated data and illustrative assumptions · results hold for this simulated world

Why Build a Simulated World?

Meridian's files cannot say who lied: a lie that works leaves no label.



15

datasets: 3 random seeds × 5 interaction strengths

50,000

applicants in each dataset

100%

of true risks known, so errors can be measured

No public dataset reveals the concealed truth, so the paired true-vs-declared design measures detection and repricing directly.

Core Research Objectives

Meridian's board wants four answers.

1. **On honest applicants, who prices risk more accurately: rules or ML?**

   HONEST APPLICANTS · ACCURACY

2. **Do the premiums cover the expected claims?**

   HONEST APPLICANTS · SOLVENCY

3. **When applicants lie, what breaks, and who catches the liars better?**

   LYING APPLICANTS · DETECTION

4. **When is checking worth its cost, and what penalty stops lying?**

   ECONOMICS · DETERRENCE

Goal: test both methods fairly and report whoever wins, not prove that ML is better.

Simulated data and illustrative assumptions · results hold for this simulated world

The Simulation Environment & 15 Datasets

To test fairly, Meridian builds fifteen worlds instead of one.

Why 15 datasets?

5 γ values × 3 seeds = 15

- One dataset can crown a method by luck
- γ: how strongly risk factors compound (real value unknown)
- Seeds: repeat 3× with fresh randomness
- Seeds disagree → flagged as unclear

50,000
applicants each

34
columns (true + declared)

≥ 88
claims in smallest group

The interaction dial (γ)

How much do risk factors make each other worse?



risks simply multiply — risks compound

5.3×
risk at γ = 0

8.4×
risk at γ = 1

Smoker + high blood pressure + high BMI vs none. Average claim rate fixed at 5.5% at every γ.

Traditional Rule-Based System (Arm A)

Meridian's status quo: a table, some multipliers and a cut-off.

Base rate by age band & sex → **× Loadings M (product)** → **× scaling k ≈ 0.53** → **Decision + premium**

Fixed loadings

Risk factor	Loading
Smoker	2.50
Heart disease	2.00
Diabetes	1.75
Six other factors*	1.25

*Hypertension, BMI ≥ 30, heavy alcohol, occupation class 4, family history, cholesterol ≥ 240

Decision rule on M

Standard
M ≤ 1.25

Rated
M ≤ 5.0

Decline
M > 5.0

- BP ≥ 140 or HbA1c ≥ 6.5 counts as the condition, even if undeclared
- Arm A+: k re-set per declared group, a fair opponent
- Loadings fixed before any result, rounded to 0.25

Machine Learning Model Framework

The innovation: let the data, not a rulebook, set the price.

GLM

Adds up risk effects, learned from data

$$\ln(p / (1 - p)) = b_0 + \Sigma\, b_j\, x_j$$

Cox

Models time until a claim

$$p_{10} = 1 - \exp(-H_0(10) \cdot e^{\text{score}})$$

XGBoost

Many small trees, the main ML model

$$\ln(p / (1 - p)) = \Sigma\, f_m(x)$$

learns more from data →

60 / 10 / 30 split by applicant: both versions of a person stay together

Trained two ways: honest records (R1) or declared records (R2)

Same columns for all methods; SHAP only describes XGBoost

Underwriting Mathematical Formulations

The four formulas that decide every price in the study.

1 · Truth: the simulated world

$$h(t) = Z \cdot A_0 \cdot s \cdot e^{bt} \cdot e^{LP}$$

$$LP = \Sigma\, \beta_k\, x_k + \gamma\, \Sigma\, \beta_{jk}\, x_j\, x_k$$

$$p(T) = 1 - (1 + 0.25 \cdot H(T))^{-4}$$

Z hidden extra risk · γ interaction dial · H(T) total risk over T years

2 · Traditional price (Arm A)

$$p_{10} = 1 - \exp(-10 \cdot k \cdot h_{band} \cdot M)$$

$$M = \Pi\ \text{loading}_j$$

Standard if $M \le 1.25$ · Rated ≤ 5.0 · else Decline

$k \approx 0.53$ fitted scaling · h_b and base rate by age band and sex

3 · Machine learning (Arm B)

GLM: $\ln(p/(1-p)) = b_0 + \Sigma\, b_j\, x_j$

XGBoost: $\ln(p/(1-p)) = \Sigma_m\, f_m(x)$

Cox: $p_{10} = 1 - \exp(-H_0(10) \cdot e^{score})$

score = $\Sigma\, b_j\, x_j$ · f_m small trees (rate 0.05, depth 2–4) · H_0 baseline risk

4 · Premium and solvency

$$P = 1.25 \times p \times SA \div T$$

Loss ratio = $\Sigma\, p \cdot SA/10 \div \Sigma\, P$

Correct book: $1 \div 1.25 = 0.80$

15% expense + 10% profit loading (assumed) · SA sum assured · T term

Simulated data and illustrative assumptions · results hold for this simulated world

The Adversarial Element: The Lying Applicant

Some applicants shade the truth, and Meridian only ever sees the form.

Share of applicants (our assumption)



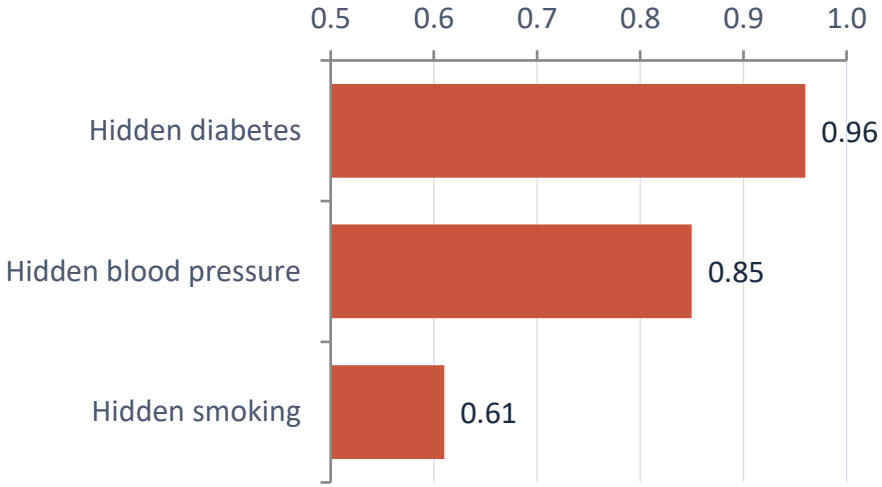
- Honest · 32,500
- Hides smoking · 7,500
- Hides disease · 6,000
- Hides both · 4,000

Lies are false values, never blanks

Only a real condition can be hidden

Smokers oversampled to 35%: about 66% of them lie

How easy a lie is to spot
(1.0 = always, 0.5 = never)

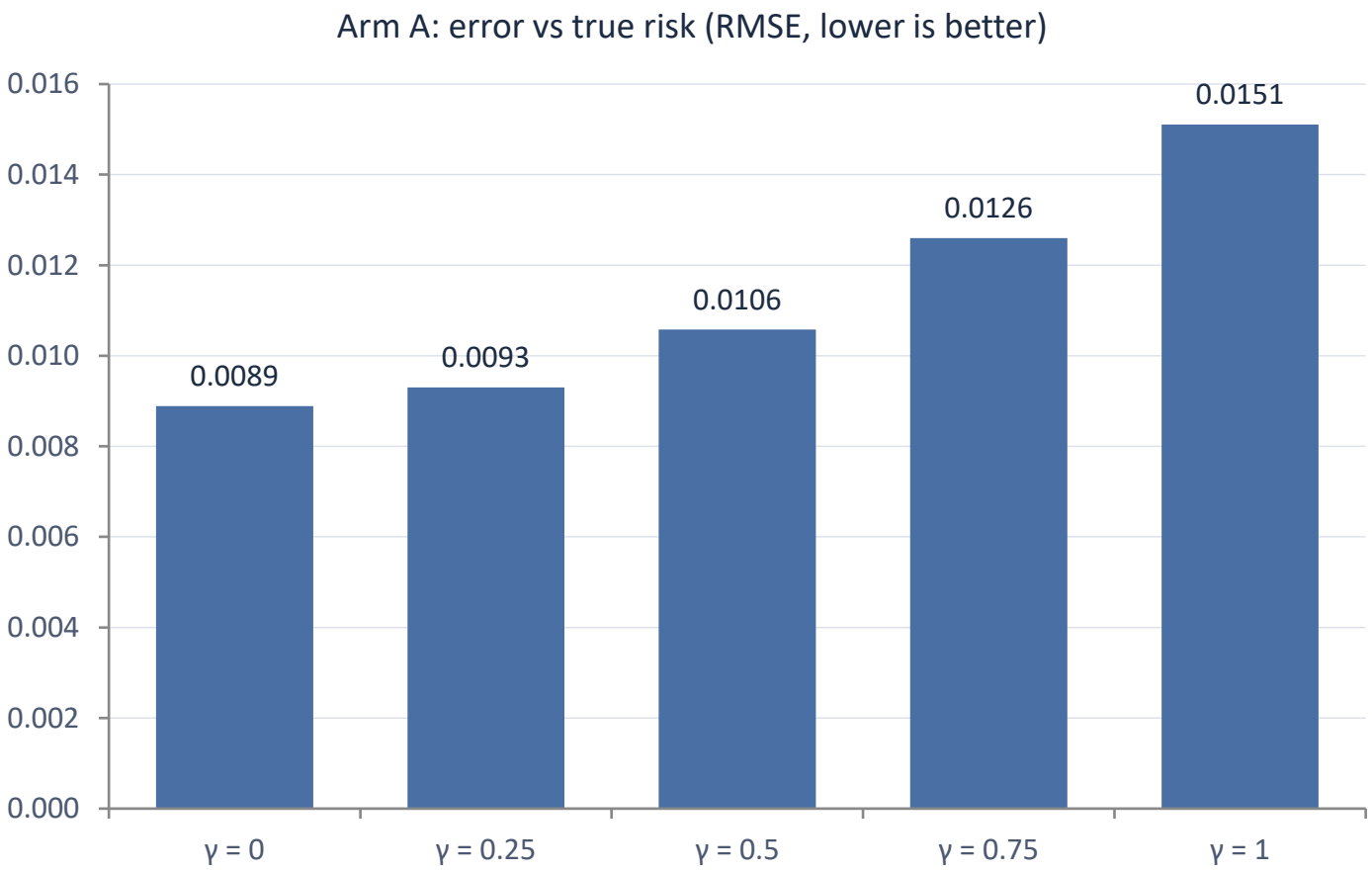


Footprints. Hidden facts leave traces in measured numbers, such as blood sugar. Smoking leaves almost none.

Simulated data and illustrative assumptions · results hold for this simulated world

Honest Applicants: Rule-Based Outcomes

The test begins with honest applicants and the status quo.



+70%

error rise from γ = 0 to γ = 1 (0.0089 → 0.0151)

Flat

does not learn: 0.0089 → 0.0088 as training grows to 270,000 (γ = 0)

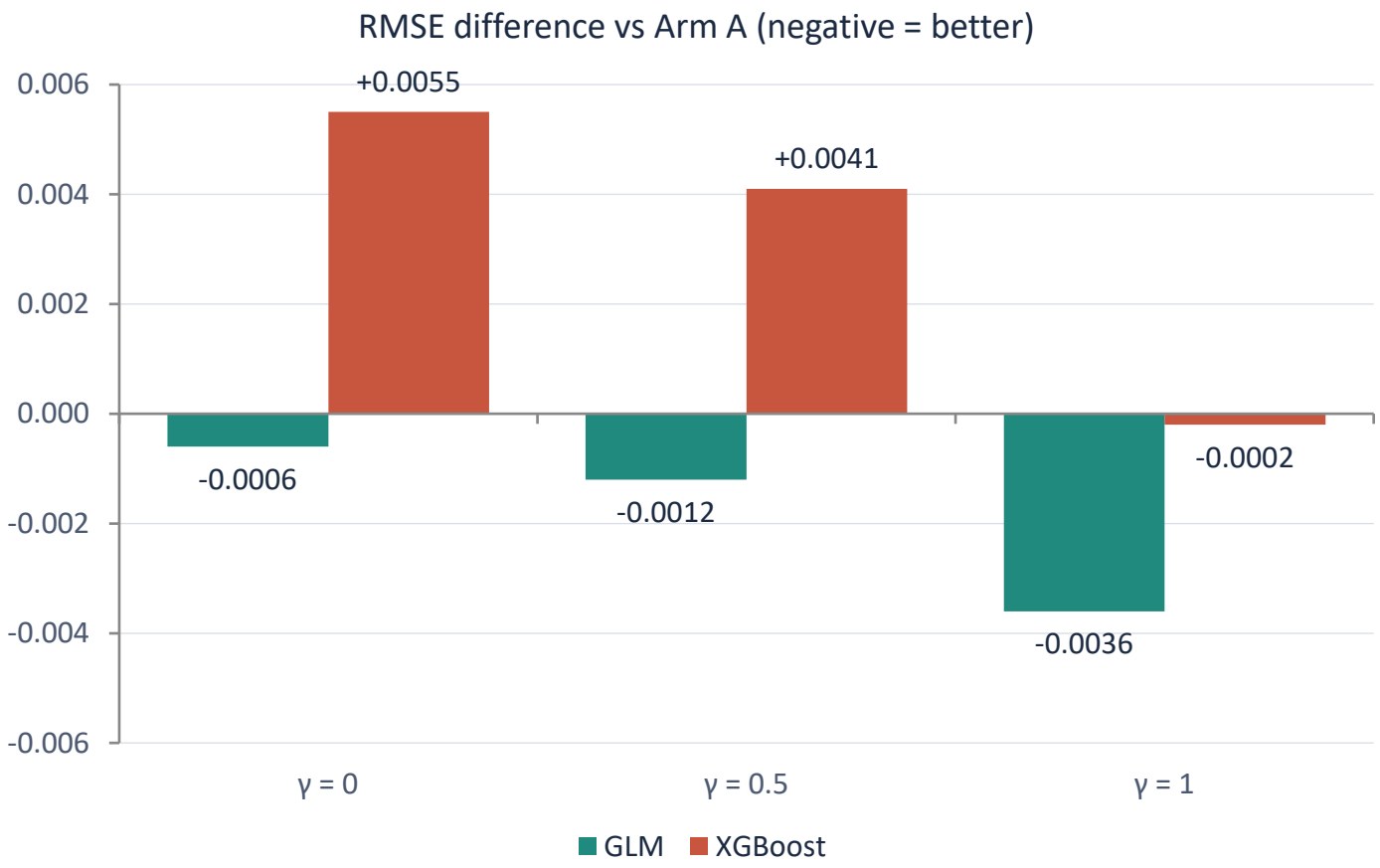
k ≈ 0.53

scaling shows the table fits the general population, not insured lives

Simulated data and illustrative assumptions · results hold for this simulated world

Honest Applicants: ML Model Outcomes

Now the challengers enter, and the story gets more complicated.



GLM and Cox slightly better at γ ≥ 0.75

All 3 seeds agree at γ = 1; unclear at low γ

XGBoost worse when γ ≤ 0.75

About equal at γ = 1 (p = 0.43)

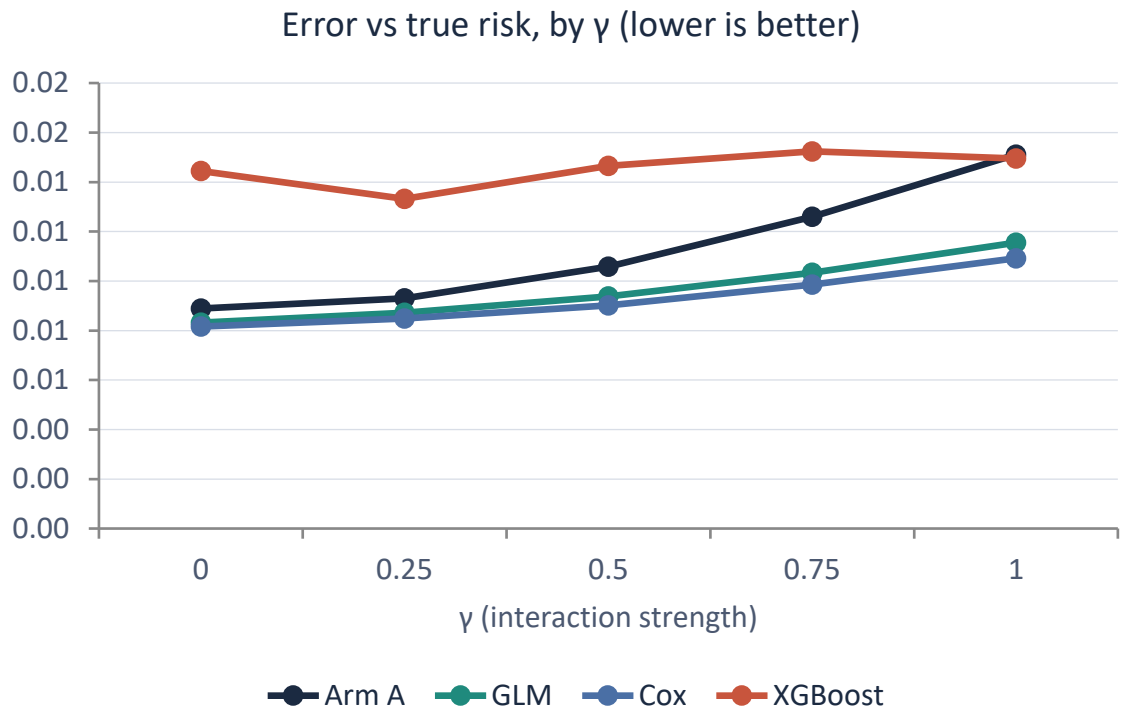
AUC cannot separate the methods

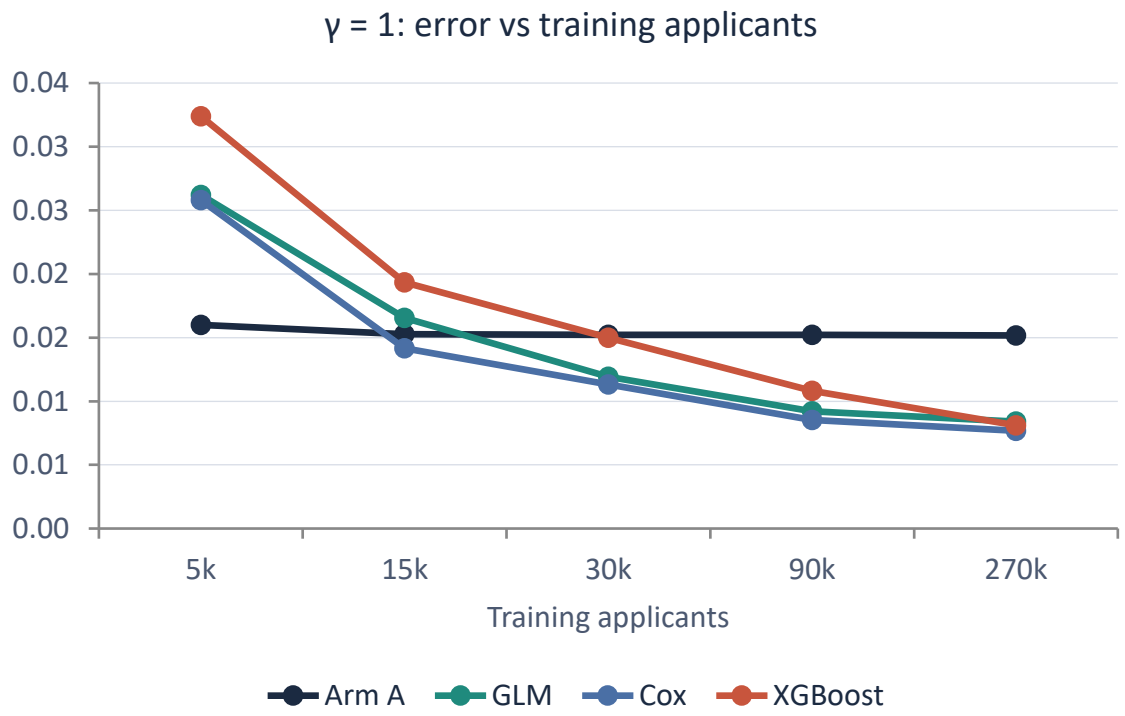
Ceiling ≈ 0.71–0.73: 10-year claims are mostly luck

Simulated data and illustrative assumptions · results hold for this simulated world

Comparative Accuracy (Objective 1)

Meridian's verdict on honest applicants: it depends on how much data it has.



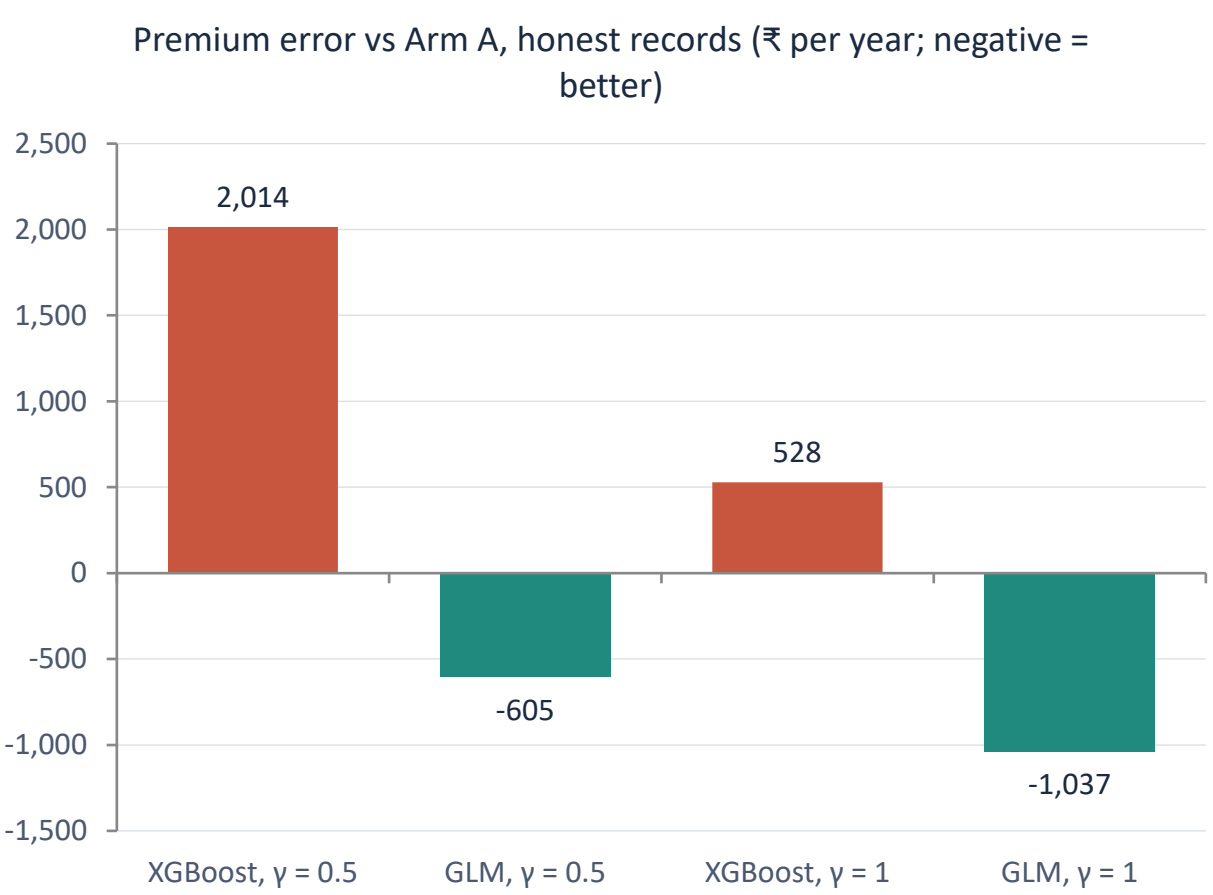


ML did not win on honest data at our size: GLM and Cox edge ahead, and XGBoost needs 30k–270k applicants.

Simulated data and illustrative assumptions · results hold for this simulated world

Premium Solvency & Claims Coverage (Objective 2)

Do Meridian's premiums actually cover its expected claims?



Loss ratio = claims ÷ premium (liars at 10%)

0.80 correct book (1 ÷ 1.25)

0.86–0.91 trained on honest records: under-charged

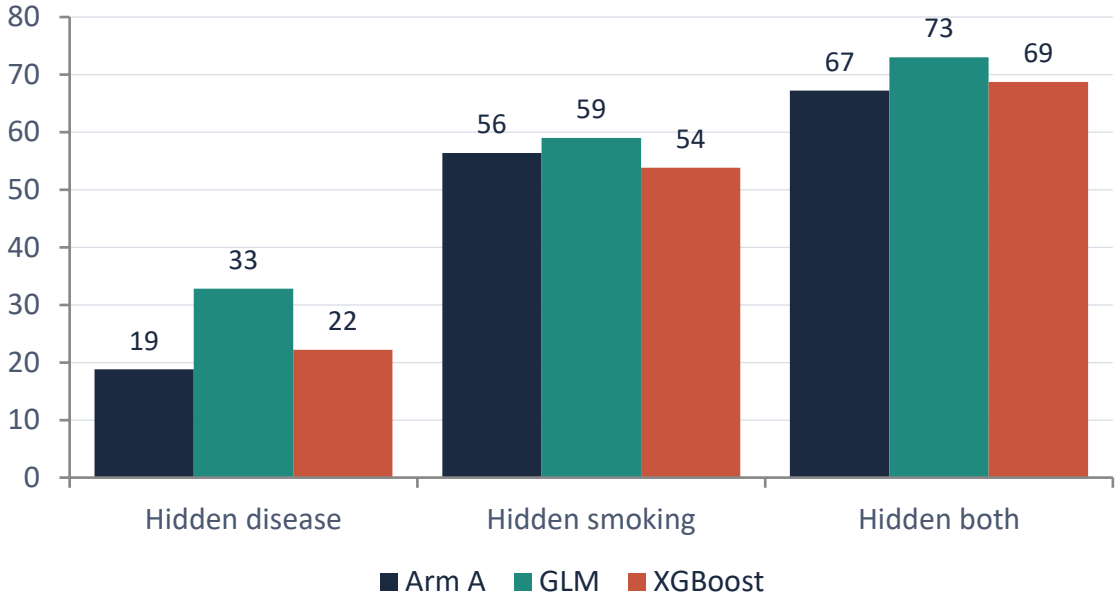
0.62–0.65 trained on declared records: over-charged

Simulated data and illustrative assumptions · results hold for this simulated world

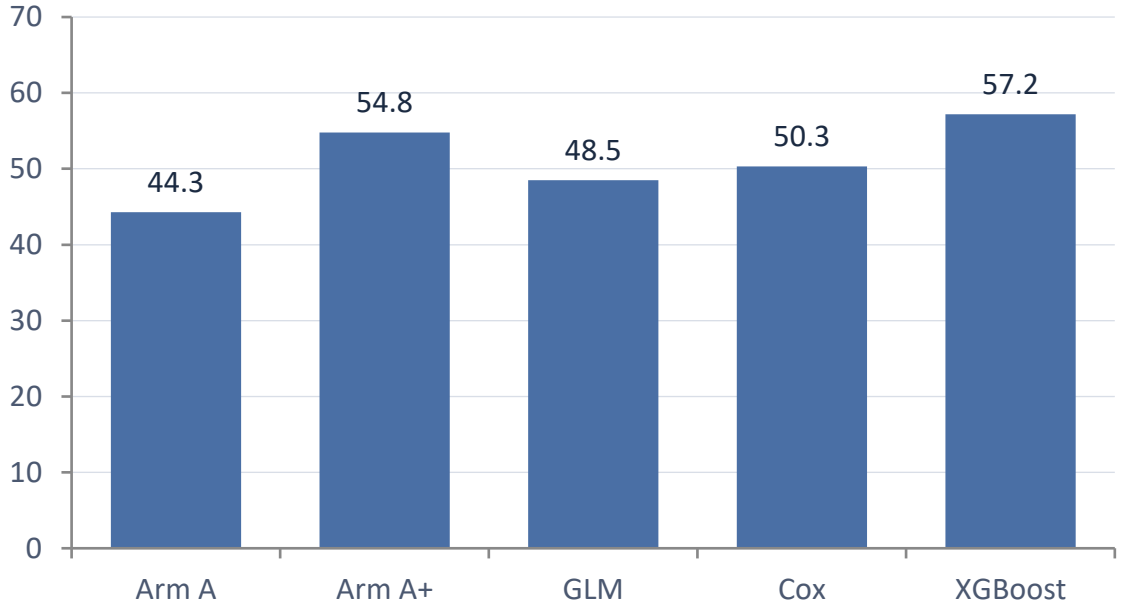
System Vulnerabilities: What Breaks When Applicants Lie?

The conflict: honest applicants versus strategic liars.

Liars underpriced (%), trained on honest records



Honest non-smokers overcharged (%), trained on declared



92.5%
of would-be declines get accepted

₹37–40k
saved a year by hiding smoking

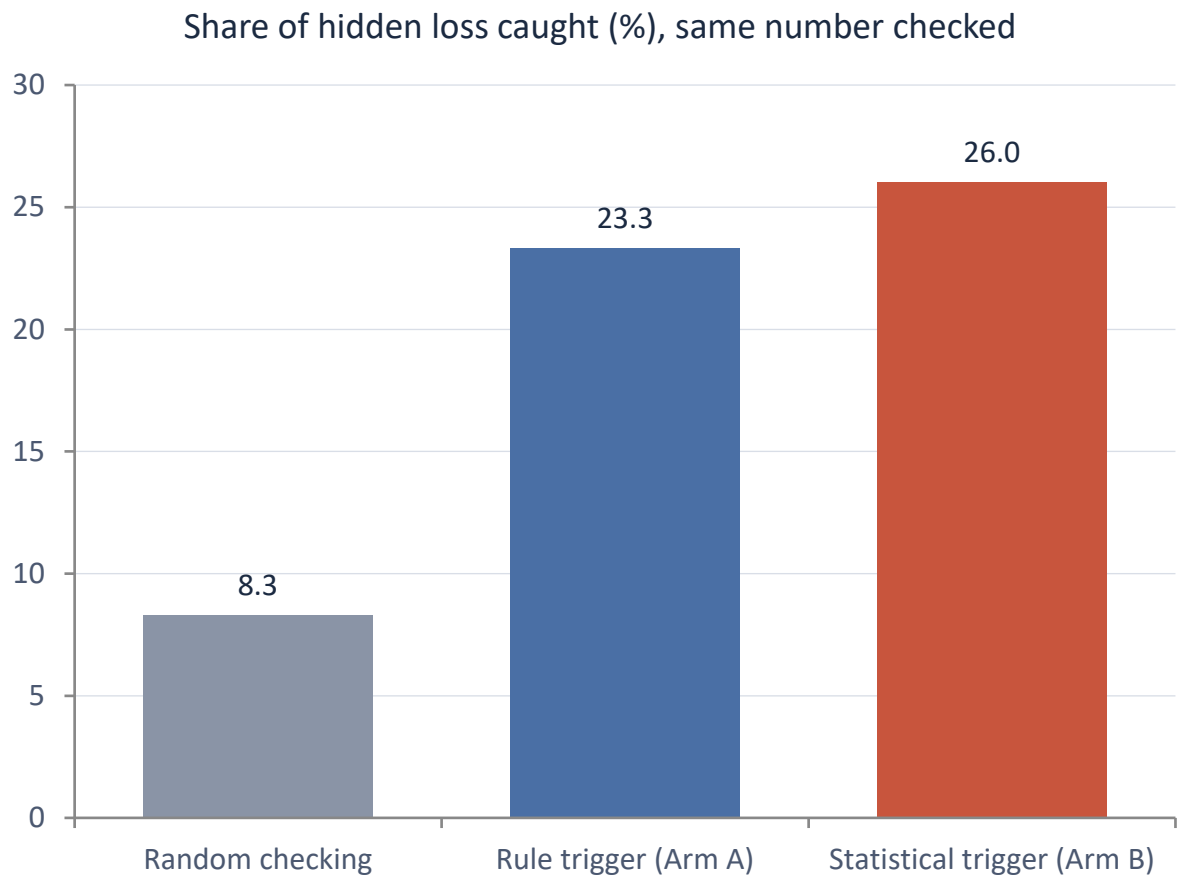
–0.003 to –0.009
error when trained on declared data

44–57%
honest non-smokers overcharged

Simulated data and illustrative assumptions · results hold for this simulated world

Fraud Detection: Who Catches the Liars Better?

Meridian can afford to check only a few applicants. Whom should it pick?



Arm A · rule trigger

Declared no high BP but reading ≥ 140; declared no diabetes but HbA1c ≥ 6.5

Arm B · statistical trigger

Consistency model trained on declared data only; never sees who lied

+2.7 pts

statistical vs rules (95% CI 1.9–3.5)

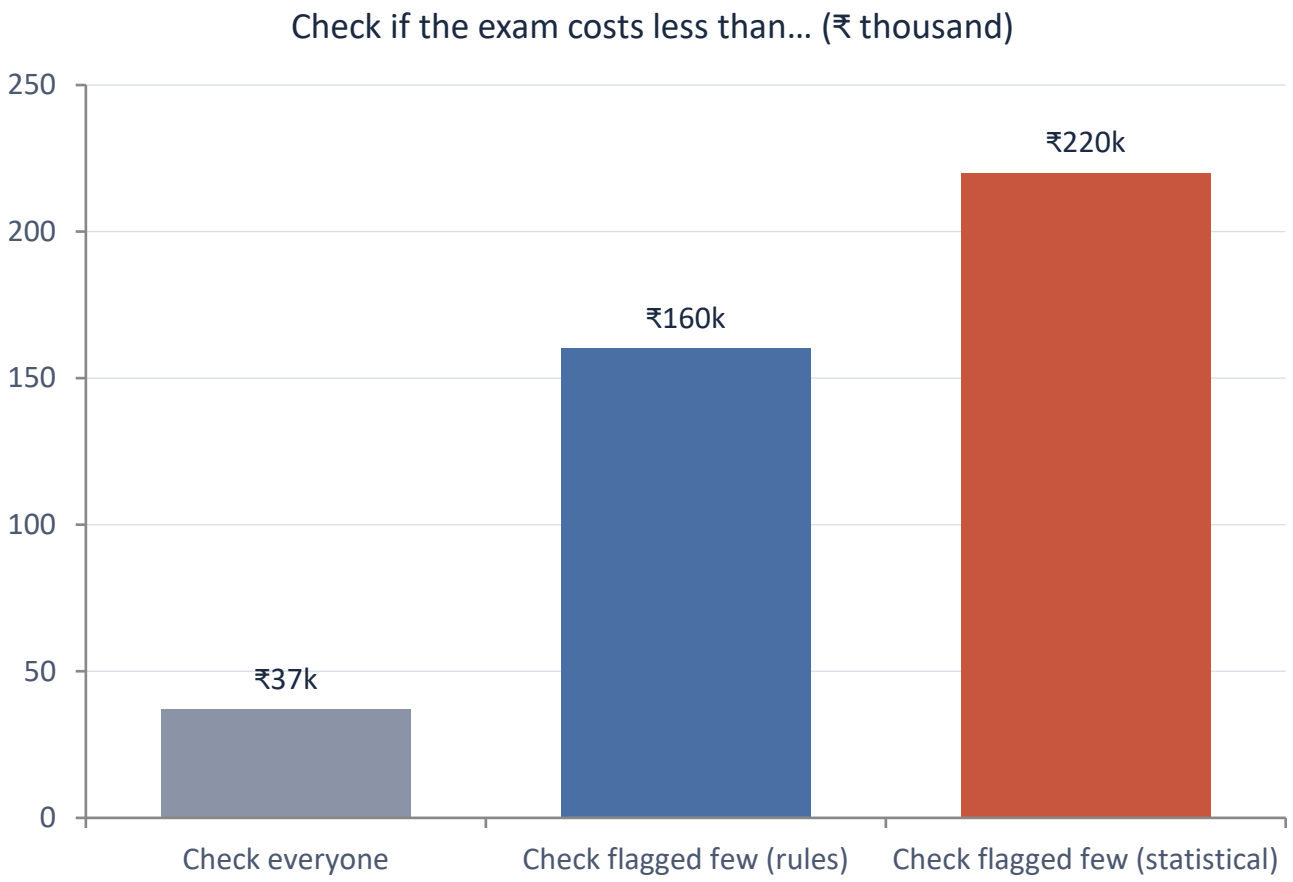
2–4%

of hidden smokers found by either tool

Simulated data and illustrative assumptions · results hold for this simulated world

The Economics of Auditing: Cost-Benefit Analysis (Objective 4)

Checking has a price. When does it pay for Meridian?



Check a flagged applicant if

$$c < P(\text{lying} \mid \text{flagged}) \times v \times d$$

c exam cost · v exam accuracy · d premium saved by lying

₹5.5k → ₹111k

break-even when checking everyone, for a ₹10 lakh → ₹2 crore policy

Assumes 10% of applicants lie, a perfect exam (v = 1) and γ = 0.5.

Simulated data and illustrative assumptions · results hold for this simulated world

Strategic Deterrence: Penalty Impact Analysis

What penalty would make lying a bad bet?

1

Lying pays if

$$G = d - \sigma \cdot r \cdot v \cdot L > 0$$

saving minus chance of being caught × penalty

2

Truth is best when

$$L \geq d \div (r \cdot v)$$

penalty at least the saving divided by detection chance

3

Insurer checks if

$$c < P(\text{lying} \mid \text{flagged}) \cdot v \cdot d$$

exam cost below the expected recovery

2.5–3×

the premium saved: penalty needed for hidden diseases or several hidden facts

Realistic

27–54×

the premium saved: penalty needed for hidden smoking

Not realistic without a better test

d premium saved · L total loss when caught · r share of liars flagged · v exam accuracy · c exam cost · σ chance of checking

Simulated data and illustrative assumptions · results hold for this simulated world

Trade-off Matrix: Rules vs Machine Learning

No method wins on every measure. Meridian must choose what it values.

Criterion	Arm A (rules)	GLM / Cox	XGBoost
Honest accuracy vs Arm A	Reference	Slightly better at every γ	Worse at γ ≤ 0.75; about equal at γ = 1
Applicants needed to pass Arm A	Never learns	15,000–30,000	30,000–90,000 (γ = 1); 90,000–270,000 (γ = 0)
Premium error vs Arm A (γ = 0.5)	Reference	–₹605 a year	+₹2,014 a year
Liars underpriced, γ = 0.5 (disease / smoking / both)	19% / 56% / 67%	33% / 59% / 73% (GLM)	22% / 54% / 69%
Trained on declared records, γ = 1	Improves 0.0036; Arm A+ is the fair reference	Improves 0.0089; about equal to Arm A+	Improves 0.0076; worse than Arm A+

A GLM is the simplest method that beats fixed rules; XGBoost needs far more data; with honest-trained models and liars, rules are hardest to beat.

Simulated data and illustrative assumptions · results hold for this simulated world

Interactive Tool Overview: The Streamlit Calculator

Meridian's underwriters can try it on one applicant at a time.

Model	Declared record	Gap to fair	True record
Traditional (Arm A)	₹20,699	−50%	₹50,470
GLM	₹17,769	−58%	₹43,503
XGBoost	₹28,063	−33%	₹52,285
Fair price (truth)	**₹41,774**		**₹41,774**

represent different pricing calculations

28% too low

the 20-year quote

The shortcut assumes constant yearly risk. True claim probability is 17.8%; the shortcut gives 12.9%.

Try it live

rdproject-calculator-rd.streamlit.app



Method	Yearly premium	Gap to true price
Traditional rules (Arm A)	₹20,699	+16.0%
Statistical model (GLM)	₹16,929	-5.1%
Machine learning (XGBoost)	₹24,359	+36.5%
True price (simulated truth)	₹17,846	-

Comprehensive Evaluation Metrics

How we measure, and how we know a difference is real.

RMSE (primary)
$\sqrt{\text{mean}\,(\text{predicted} - \text{true p})^2}$
Average error vs true risk; lower is better

Premium error
mean | predicted − true premium |
In ₹ per year

Underpricing %
mean 100 × (p − predicted) ÷ p
Positive = liar charged too little

Loss ratio
Σ p · SA/10 ÷ Σ premium
Correct book = 0.80

AUC
Pr(claimant ranked above non-claimant)
Ranking power; ceiling ≈ 0.71–0.73

Difference-in-differences
gap on declared − gap on true
Positive = ML edge grows when applicants lie

Bootstrap interval
1,000 resamples by applicant
95% interval = middle 95%; paired design

Holm correction
adjusts p across many tests
Prevents false discoveries

DeLong test
z = ΔAUC ÷ SE
Compares two AUCs on the same applicants

Simulated data and illustrative assumptions · results hold for this simulated world

Deep Dive into Metric Discrepancies

Why the numbers seem to disagree, and what Meridian should read into them.

0.71–0.73

AUC ceiling

AUC cannot separate methods: only ~5.5% claim in 10 years, so outcomes are mostly luck. RMSE vs true risk is primary.

35% vs 10%

Liar share

44–57% overcharge in our cohort; at a realistic 10% liars the loss ratio is 0.62–0.65 vs a correct 0.80.

1 → 3 of 3

Seeds agreeing

GLM beats Arm A in 1, 2 and 3 of 3 seeds at γ = 0, 0.5, 1. Where seeds disagree we say unclear.

3 seeds

Narrow intervals

Intervals treat trained models as fixed, so true uncertainty is larger than shown.

Research Limitations

Every honest study states what it cannot claim.

Simulated world

Not real insurers. Interaction sizes and liar shares are assumptions.

Simplified traditional engine

Simple loadings on a general-population table; no SOA insured-lives table.

Smoker oversampling

35% smokers vs ~11–14% in reality; re-weighting fixes only the liar share.

Only 3 seeds

Intervals ignore training uncertainty, so they are too narrow.

Missing dynamics

No lapses, discounting, repeated play or insurer reaction.

Scope limits

No fairness test beyond age, sex, risk band; no ML running cost; 20–30-year quotes carry large error.

Future Scope & Roadmap

What Meridian's next study should do.

1 Real data

- Real insured-lives tables (SOA VBT, Indian assured-lives)
- Real applicant data
- Nonlinear risk generator as a sensitivity check

2 Stronger statistics

- More seeds
- Training uncertainty inside the intervals

3 Wider scope

- Fairness across protected groups
- Running and compliance cost of ML
- Lapses, discounting, repeated play

4 Detection & product

- Nicotine or biomarker test for hidden smoking
- Term-aware calculator with a time-varying hazard



Biggest open gap: hidden smoking, found in only 2–4% of cases by either tool.

Strategic Recommendations for the Insurance Industry

What Meridian Life should do on Monday morning.

1	**Start with a GLM, not a complex model**	At ~30,000 training applicants the GLM beat fixed rules; XGBoost needed 90,000+ to catch up.
2	**Train on declared data, then audit for overcharging**	Error falls 0.003–0.009, but honest non-smokers pay 44–57% more (at 35% liars). Benchmark against Arm A+.
3	**Cross-check declarations with measured readings**	BP and blood-sugar triggers catch ~20% of liars; a statistical trigger adds +2.7 points of hidden loss.
4	**Verify the flagged few, not everyone**	Checking everyone pays below ≈ ₹37k per exam; checking the flagged few pays up to ₹1.6–2.2 lakh.
5	**Fix the smoking blind spot**	Only 2–4% of hidden smokers are found; deterrence needs a penalty of 27–54× the saving. Add a nicotine test.

Simulated data and illustrative assumptions · results hold for this simulated world

Conclusion

The verdict: do not buy complexity. Buy data, verification and a better smoking test.

1

Learning from data beats fixed rules once there is enough data

2

Complex ML added nothing over a plain GLM at our data size

3

Lies move prices more than the choice of method

4

Checking pays at real exam costs; smarter flagging stretches the range

Machine learning did not clearly win. These conclusions hold for this simulated world only, not for real insurers.

Q&A and References

Meridian's story ends here. Your questions begin.

Questions?

Try the calculator:

rdproject-calculator-rd.streamlit.app

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Supervisor: Prof. Shaishavi Sabnis

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